

PRANESH V

+91 8807149866 esppranesh@gmail.com
in linkedin.com/in/pranesh-v-0451b0314 github.com/White-fang10

Professional Summary

Motivated and versatile Information Technology undergraduate in the final year of study, with hands-on experience in full-stack development, embedded systems, and intelligent application design. Proven ability to lead and deliver end-to-end software solutions spanning web platforms, real-time monitoring systems, and AI-integrated tools. Adept at translating complex problem statements into scalable, user-centric products. Strong foundation in system design, data structures, and cross-functional collaboration.

Education

V.S.B Engineering College

2023 – 2027

B.Tech in Information Technology, Karur, Tamil Nadu

- Currently pursuing 4th year; CGPA: 7.6
- Relevant coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Web Technologies

Achievements

- **Patent Filed (2024):** Invented and filed an Indian patent for a **Space Debris Management System** — a computational model leveraging orbital tracking algorithms and predictive analytics to identify, catalogue, and mitigate the risk posed by space debris to active satellites and future space missions.
- **Patent Filed (2025):** Filed an Indian patent for **Hawk AI — Automated Attendance System**, a deep learning-based attendance management solution that leverages smart board cameras for real-time facial recognition, eliminating the need for manual roll calls in educational and corporate environments.
- **NPTEL Certifications:** Completed NPTEL-certified courses in *Java Programming* and *Big Data Analysis*, demonstrating proficiency in industry-relevant technologies.
- **Board Certification:** Completed *Data Science for Beginners* certification, reinforcing foundational knowledge in data analysis and visualization workflows.

Experience

Full Stack Development Intern

2024

Software Development Internship

Remote / India

- Developed and deployed web application modules using modern frontend and backend technologies, gaining practical exposure to the complete software development lifecycle.
- Collaborated with cross-functional teams to design RESTful APIs, integrate third-party services, and ensure responsive UI/UX across devices.
- Contributed to code reviews, documentation, and iterative feature development following Agile methodologies.

Projects

Hawk AI | *Deep Learning, Computer Vision, Python, OpenCV*

- Designed and developed an **automated attendance management system** powered by deep learning and facial recognition, operating exclusively through existing smart board cameras — requiring zero additional hardware installation.
- Implemented real-time face detection and identification pipeline using convolutional neural networks, achieving high accuracy under varied lighting and classroom conditions.
- Generated automated attendance reports and integrated the system with institutional databases, significantly reducing administrative overhead and human error.

House | *React, Node.js, WebSockets, Judge0 API, CSS Animations*

- Built a **Harry Potter-themed online competitive coding platform** featuring a fully immersive, dynamically themed user interface with house-based team selection, animated environments, and lore-driven UX copy.
- Integrated a real-time code execution engine supporting multiple programming languages, with live contest timers, leaderboards, and problem difficulty tiers.
- Designed a responsive frontend with custom CSS animations and transitions to enhance user engagement and deliver a distinctive competitive coding experience.

Investigator | *React, Node.js, Live Market APIs, AI Integration, Chart.js*

- Developed a comprehensive **investment monitoring and simulation platform** enabling users to track multi-asset portfolios, monitor real-time market prices, and analyze historical performance through interactive dashboards.
- Built an **AI-powered investment mentor** that provides contextual insights, portfolio rebalancing suggestions, and risk assessments based on the user's current holdings and market conditions.
- Implemented an investment simulator allowing users to run hypothetical investment scenarios against live market data, facilitating informed and data-driven financial decision-making.

ClarifAI | *RAG, Vector Database, Endee, Python, Full Stack, LLM*

- Built a **production-ready intelligent RAG (Retrieval-Augmented Generation) platform** designed as an AI assistant for events, hackathons, and conferences, enabling high-accuracy, hallucination-free information retrieval with full source attribution.
- Leveraged the **Endee vector database** to index and semantically search event documents — rulebooks, schedules, and coordinator uploads — allowing participants to query complex information in natural language with precision.
- Architected a full-stack event management interface for coordinators to upload and manage documents, and a participant-facing query interface delivering contextually grounded answers with cited references.

Project Shield — LT Line Breakage Detection System | *ESP32, ML, LoRa, GSM, Cloud*

- Led the end-to-end development of an **intelligent disaster management platform** for detecting low-tension electrical line failures using ESP32 IoT sensors and machine learning-based anomaly detection algorithms.
- Built a cloud-hosted real-time dashboard with automated GSM and LoRa alert dispatching and remote circuit isolation capabilities, minimising response time during critical failure events.
- Coordinated system architecture design, hardware integration, and firmware development as Lead Developer, delivering a functional prototype for rural grid infrastructure safety.

Technical Skills

Languages: Python, Java, JavaScript, C, SQL

Web & Frameworks: React.js, Node.js, Express.js, HTML5, CSS3, REST APIs

Tools & Platforms: Git, GitHub, VS Code, Postman, Linux, Arduino IDE

Domains: Full Stack Development, Embedded Systems, Computer Vision, Data Structures & Algorithms

Soft Skills: Leadership, Creative Problem Solving, Communication, Adaptability, Team Collaboration